

Work Smarter with Microsoft's Cutting-Edge AI



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*Data & Applied Scientist II
Microsoft*

A little about me...

- **Microsoft | Bing Autosuggest** (2021 – Present)
 - Data & Applied Scientist II (2023 - Present)
 - Data & Applied Scientist (2021 - 2023)
- **IIT Bombay** (2017 – 21)
 - Major in Mechanical Engineering (9.90 CPI)
 - Minor in Computer Science & Engineering
 - Department Rank 1 | Institute Rank 2
- **Amazon #1 Best Selling Author** of “Beyond Code”
- Holder of **1 US Patent**
- **Areas of Interest:** AI Tools, NLP, Data Analysis



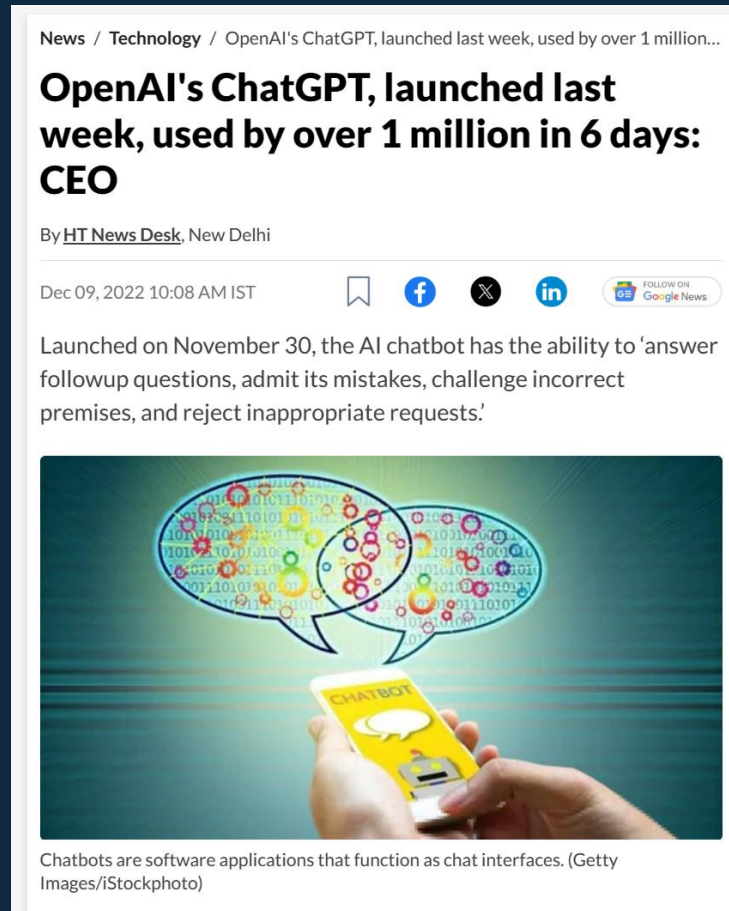
Agenda

- The AI Revolution & Microsoft's Role
- Leverage **Bing Chat** for (almost) any Task
- Streamline Research with **Edge Copilot**
- Create Stunning Image with **Bing Image Creator**
- Design Creatives with **Microsoft Designer**
- Code Smarter with **GitHub Copilot**
- Q&A

The AI Revolution & Microsoft's Role



April 2022



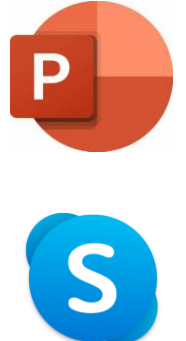
December 2022



February 2023

... since then, AI has been dominating the industry, changing every paradigm of work with new Models, Tools & Capabilities almost every week!

The Suite of Microsoft Copilots

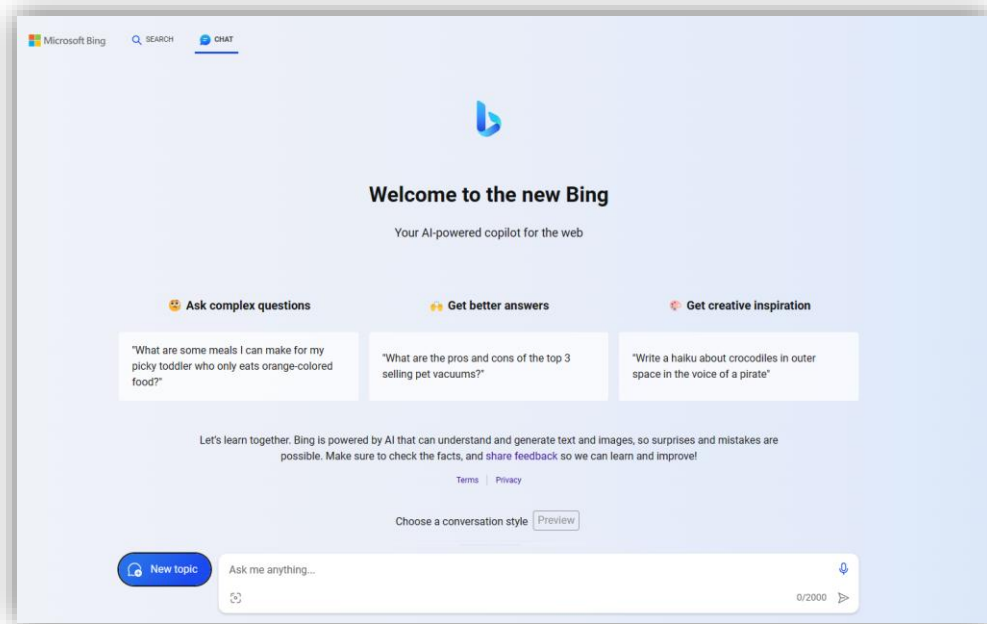




Leverage Bing Chat for (almost) any Task

What's Bing Chat & How Can I Get Started with It?

- AI-Powered Chat integrated with Bing Search



<https://www.bing.com/chat>

- **Why Bing Chat?**
 - Integrated with Bing Search Engine – provides more accurate & relevant answers
 - Access to other features like Bing Image Creator & Image Inputs
 - Available across platforms
 - ***FREE Access to GPT-4***

Using Bing Chat, the Right Way

Are you NOT getting desired results with Bing Chat?

Maybe...

You are falling short of writing high-quality prompts that give you awesome results!



Created by Tezan, using AI

Unlock the True potential of Bing Chat using High-Quality Prompt Templates

- Sample Templates to Complete Time-Consuming Tasks (demo)
- Best Practices for writing High-Quality Prompts
- Generic Framework to Create your Own Templates for Any Task

Sample Prompt Template 1: Ideation

Prompt 1

You are a seasoned consultant in **[domain of interest]** and an expert with 20 years of experience. You are creative and knowledgeable about this topic and ideally have practical experience in it as well.

I ask you to focus on one specific problem to solve which is: **[topic of interest]**.

Look at the topic from an innovative and non-traditional perspective in order to come up with out-of-the-box solutions.

Then analyze any ideas that you come up with to ensure they are promising and then prioritize them.

Develop actionable steps for the plan and methods for evaluating its implementation and quality.

Complete the above steps to generate 2 top ideas.

Prompt 2

Idea 1: **[Describe Idea 1]**

Idea 2: **[Describe Idea 2]**

Write an analysis of the 2 ideas based on the following criteria: **[Specify your criteria, like market size, customer demand, profitability, competition, innovation, etc.]**

- Use a tabular format to present the analysis.
- Use markdown syntax to create the table.
- Use bullet points to list the pros and cons of each idea for each criterion.
- Use a rating scale of 1 to 5 stars (★) to indicate how well each idea performs on each criterion.
- Use bold text to highlight the overall winner for each criterion

Sample Prompt Template 2: Seek Technical Support

Prompt

You are a certified **[software]** expert, with 10+ years of experience using it and training people to use it.

I need to accomplish the task given within the <task> </task> tags using **[software]**. I need you to guide me to complete this task with detailed step-by-step instructions, and code (wherever necessary).

<task>

[Describe your task]

</task>

Help me accomplish this task in the fastest way possible, with the least effort.

Prompt Engineering Best Practices

- Declare a **role** (expert persona)
 - Roleplay-based prompts work really well, according to my observation
- State the main task clearly, with **detailed description**
 - The more details you provide, the better results you can expect
- For tasks involving multiple steps, give it "**time to think**"
 - Break the original task into multiple prompts, chained together
- While supplying additional context, use "**tags**" & other identifiers
 - Like <xyz></xyz>, etc.
- Supply **constraints** explicitly
 - Word limit, time involved, etc.
- Start a **New Task** in a **New Chat**
 - To prevent unnecessary context from previous chat

RACEF Prompt Framework

Role

You are **[accomplished persona]**, specializing in **[broad category/domain which your desired task falls under]**.

Action

[Mention the context that will be provided]. Given this, your task is to **[description of the task]** to achieve **[the objective of the task]**.

[Describe the detailed steps to accomplish the task]

Context

[Additional context provided in relevant tags]

Examples

[Give samples of outputs]

Format

[Define the constraints on the output]

Write in the style of **[tone of voice]**, targeted towards **[audience]**.



Streamline Research with Edge Copilot

Streamlining Research with AI

Now, you can extract desired answers from large documents without having to read through the entire document!



Created by Tezan, using AI

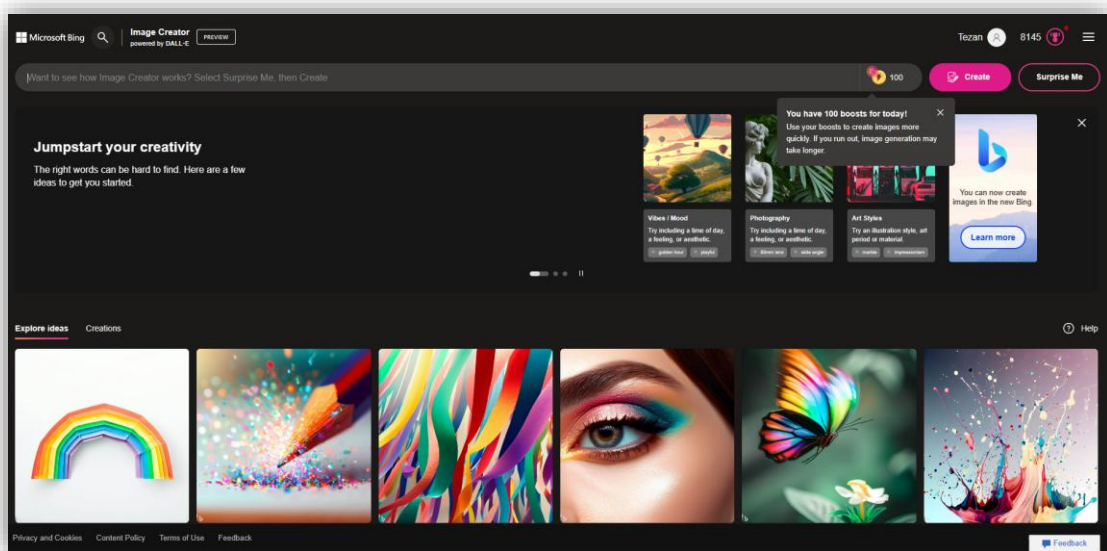
Use the Edge Copilot & start a conversation with your documents...

Image Creator
powered by DALL·E

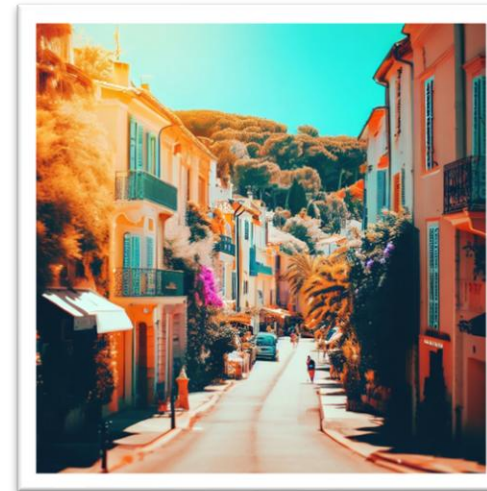
Create Stunning Image with Bing Image Creator

Bring Your Imagination to Life with Image Creator

- Generative AI Art Tool
 - Create stunning images based on text
- Powered by Dall E
- 100 boosts/day on signup
- Perfect for:
 - Newsletters
 - Blogs
 - Presentations
 - ...and much more



<https://www.bing.com/create>



(Realistic) streets of Southern France, summertime, bright and beautiful, cinematic landscape



(Crazy Concepts) An astronaut surfing on the moon with Earth in the background

Its Demo Time Again!

Concept Art



a cute robot scuba diving near a coral reef, colorful & vibrant underwater creatures & plants, digital art

Photorealistic Images



royal Bengal tiger walking in a forest

Generating Realistic Images

Prompt Template

[Subject] in the ***[style/technique]***, set in a ***[environment/setting]*** with a ***[mood/atmosphere]***. Include ***[details/modifiers]***, emulate a ***[specific camera]*** and ***[aperture]***, use ***[lighting conditions]***, and compose with ***[composition/perspective guidelines]***.

But how do I get all this
technical information & jargon?

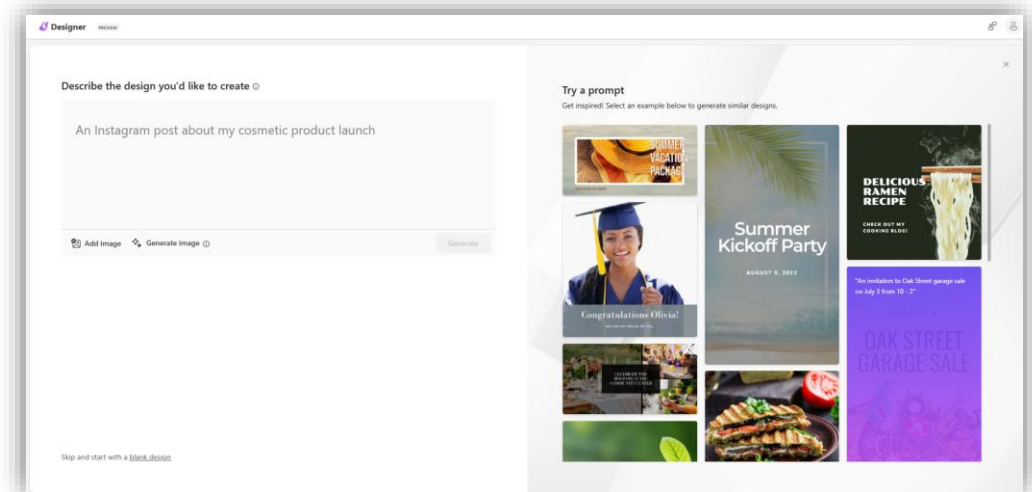
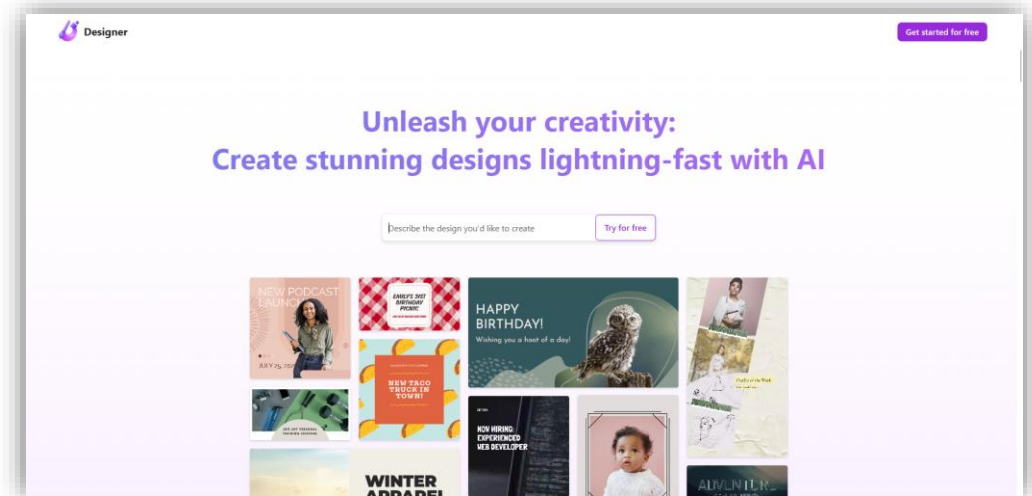
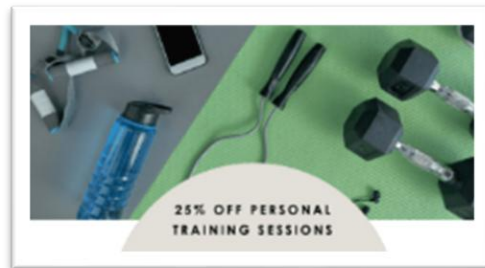
Well...leave that to Bing Chat!



Design Creatives with Microsoft Designer

Design Professional, High-Quality Creatives with AI

- Create professional-grade social media posts, invitations, digital postcards, graphics, and more
- Sign up for FREE & start using the Generative AI capabilities



<https://designer.microsoft.com/>

Let's Try Creating Some AI Creatives

Instagram Story



Create an Instagram story inviting people to attend the Ganesh Chaturthi celebrations with vibrant colors with motion

Event Poster



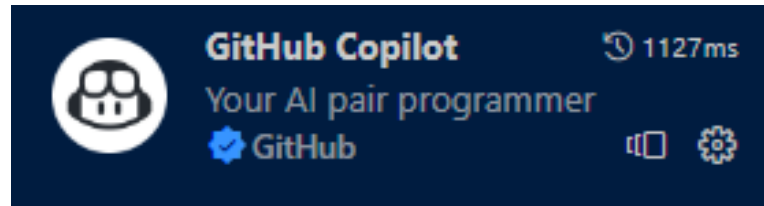
Create a modern landscape poster for an AI & Automation event in Mumbai, showcasing new technologies.



Code Smarter with GitHub Copilot

Introducing Your AI Pair Programmer – GitHub Copilot

- Available as VS Code Extension



- Works with multiple languages
- Draws context from comments & code to suggest lines as well as whole functions



GitHub Copilot X

- Features
 - Ask questions about Code
 - Analyse & Debug Code
 - Document code automatically
 - AI-Generated Descriptions for Pull Requests
 - Automated Testing of Code
- Available as VS Extension

GitHub Copilot is available under FREE Trial for 30 days, post which you must upgrade to a paid plan for continued usage

Copilot X Can be accessed by joining the waitlist

Time for Some Live Action!

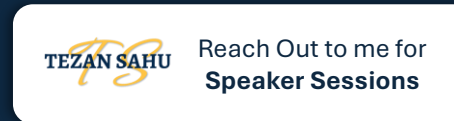
- Write a function using Copilot simply by stating the functionality
- Fix issues using Copilot
- See Copilot X in action for:
 - Code Explanation
 - Optimization
 - Documentation
 - Testing
- Learn how to build projects with AI in seconds, ***without Copilot***

Summarizing the Learnings...

- The AI Revolution that's underway
- Range of AI Copilots offered by Microsoft
- Leverage Bing Chat for various Tasks
 - Prompt Engineering Best Practices
 - RACEF Prompt Framework
- Create stunning visuals with Bing Image Creator
 - Both Concept Art & Realistic Images
 - Prompt framework for Photorealistic Image Generation
- Design creatives for a range of canvases using Microsoft Designer
- Code Smartly using GitHub Copilot & Copilot X
 - How to spin up entire projects with AI

That's All Folks!

For More Interesting Stuff on AI , Data Science & the Latest Technologies...



Comment on my LinkedIn Post with your most valuable learnings & drop in your email IDs to get the resources used in this session